

AHMED FERGANEY

Operational Excellence Engineer

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PROFILE

Operational Excellence Engineer with a proven record of driving measurable performance gains across automotive and home appliance manufacturing lines. Skilled in Lean Six Sigma, work standardization, line balancing, and KPI-driven continuous improvement, with experience leading cross-functional transformation initiatives that scale knowledge, improve line efficiency, and embed a sustainable culture of operational excellence.

EDUCATION

B.Sc. in Mechanical Engineering | Banha University | 2016 – 2022

- Academic Standing: Ranked 5th out of 120 in the Mechanical Engineering Department
- Graduation Project: Design, Modeling, and Control of a Six-DOF Artificial Upper Limb

PROFESSIONAL EXPERIENCE

Product Development Engineer | Kandil Glass | 10th of Ramadan City, Egypt06/2025 – Present

- Led end-to-end development of customized glass products, from initial client requirements (via sales) through feasibility analysis, design, and full-scale manufacturing.
- Managed cross-functional execution across design, production, and operations; monitored production performance and analyzed data to optimize efficiency and delivery.

Operational Excellence Engineer | Midea Electric | 6th of October City, Egypt02/2025 – 05/2026

- Led a work standardization (Standard Work) project across production lines, improving Line of Balance (LOB) from 60% to 75% — a 15-point gain in line efficiency and workload distribution — within the Home Appliances sector.
- Directly supervised and coached a production/process team, conducting training and performance reviews to build capability and sustain performance gains.
- Led cross-functional training and process transformation initiatives, enabling knowledge scaling and collaboration across engineering, production, and quality teams.

Process Quality Engineer | Methode Electronics | Nasr City, Egypt12/2023 – 12/2024

- Reduced defect rate from 8% to 2% (75% reduction) on the VW automotive assembly line by applying root cause analysis (RCA), PFMEA, and corrective/preventive actions (CAPA) — directly improving line throughput and quality-driven downtime.
- Applied structured problem-solving methodologies (DMAIC, DFMEA, 5 Why, Fishbone) to diagnose recurring failures and verify effectiveness of corrective actions, strengthening overall line stability.
- Coordinated with internal and external suppliers on defect containment and proactive prevention plans to reduce recurrence and protect line performance.

Continuous Improvement Engineer | JAC | Badr City, Egypt03/2023 – 11/2023

- Increased Water Dispenser line productivity from 150 to 225 units per 2-hour cycle (50% improvement) through process and workflow optimization, within the Electrical Appliances sector.
- Held continuous improvement responsibility for the Water Dispenser and TV production lines, identifying and closing efficiency gaps.

CORE QUALIFICATIONS

Continuous Improvement & Lean: Lean Manufacturing (Kaizen, 5S, SMED, Waste Reduction), Six Sigma fundamentals (DMAIC), Work Standardization & SOP Development, Line Balancing, Capacity Planning & Takt Time Analysis, Time & Motion Studies, Cycle Time Reduction

KPI & Performance Management: OEE, FPY, Scrap Rate, Downtime Reduction, Line of Balance (LOB) Improvement, Cost Reduction & Productivity Initiatives

Process Engineering: Process Design, Mapping & Optimization (VSM, SIPOC, Process Flow Diagrams), PFMEA/DFMEA & Control Plans, Root Cause Analysis (5 Whys, Fishbone/Ishikawa)

Quality Foundations: 8D, CAPA, RCA, Pareto Analysis, Audit Participation & Compliance Processes

Data & Analysis: Python for data analysis, SQL for production data querying, Statistical Process Control (SPC, Cp/Cpk, control charts, hypothesis testing), Power BI dashboards & data visualization

Leadership: Cross-functional Team Leadership, Team Supervision & Coaching, Training & Performance Reviews, Agile/Scrum Methodologies, Technical Mentoring, Problem-Solving Under Constraints

CERTIFICATIONS

- Oxyacetylene and ARC-Welding in High-Pressure Pipes Diploma | Don Bosco Institute | 2019
- Turning Course | Don Bosco Institute | 2020
- Automation Basic Diploma | HA Consulting Group | 2020
- Robotics Camp with a Case Study (Robotic Arm) | RTE | 2020
- Mechanics of Automotive Diploma | Don Bosco Institute | 2021
- Lean Six Sigma Yellow and Green Belt | Udemy | 2022
- CPIM Part One | HPA | 2023
- Embedded System Diploma | Eng. Ahmed Abd El-Ghafar | 2023
- Statistics, Linear Algebra & Numerical Optimization | ITI | 2024
- Databases, Power BI, Python for Data Analysis | Udemy | 2025
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, AI Agents | Udemy | 2026

LANGUAGES & TOOLS

Languages: Arabic (Native), English (Professional), German (Basic)

Tools: AutoCAD, Autodesk Inventor, Power BI, MS Office, Jira, Trello, MATLAB